

LUXÉ MERCHANDISE PLATFORM

INTRAKRAFT ASSIGNMENT — SETUP & RUN INSTRUCTIONS

Version: 1.0.0 | Full-Stack Assessment

1. Executive Summary & Live Deployments

This project is a production-grade apparel merchandising system engineered for Intrakraft. It strictly separates the Frontend and Backend architectures, supporting Excel catalogue ingestion, dynamic size matrixing, grade-wise cart allocation, and multi-level size ratio configuration using the Hamilton Method.

Component	Live Production URL / Repository	Environment
Live Frontend	https://lokesh314-git.github.io/Intrakraft/	GitHub Pages (SPA)
Live Backend API	https://intrakraft-backend.vercel.app	Vercel Serverless
Interactive Swagger Docs	https://intrakraft-backend.vercel.app/api/docs	OpenAPI 3.0 UI
Health Check Endpoint	https://intrakraft-backend.vercel.app/api/health	JSON Health Diagnostic
Frontend GitHub Repo	https://github.com/Lokesh314-git/Intrakraft	Main Branch & gh-pages
Backend GitHub Repo	https://github.com/Lokesh314-git/Intrakraft-backend	Production TypeScript

2. Technology Stack & System Prerequisites

Layer	Technologies Used	Key Responsibilities
Frontend	React 18, TypeScript, Tailwind CSS, Zustand, Vite, HashRouter	UI Rendering, Client State, Routing, Cart Matrix, User Views
Backend	Node.js, Express, TypeScript, SheetJS (xlsx), Multer, Swagger UI	Excel Ingestion, Data Validation, Ratio Calculations, REST API
Authentication	Firebase Authentication & Firebase Admin SDK	Google Sign-In, Session Token Validation, Role Guards
Prerequisites	Node.js v18.0+ (v20 recommended), npm v9.0+, Git installed	Required runtime dependencies to build and run locally

3. Step-by-Step Local Setup & Run Instructions

Option A: Running Both Frontend & Backend Concurrently (Recommended)

```
# 1. Clone repository and install frontend dependencies
git clone https://github.com/Lokesh314-git/Intrakraft.git
cd Intrakraft && npm install
# 2. Install backend dependencies and launch concurrent dev environment
cd backend && npm install && cd ..
npm run dev
```

- Frontend UI will launch on: `http://localhost:3000` (or `http://localhost:5173`)
- Backend REST API will launch on: `http://localhost:5000/api`
- Local Swagger Documentation available on: `http://localhost:5000/api/docs`

Option B: Running Backend & Frontend Independently

Terminal 1 (Backend Server):

```
cd backend
npm install
npm run dev

# Server running on port 5000
# Health: http://localhost:5000/api/health
```

Terminal 2 (Frontend Client):

```
cd .. (project root)
npm install
npm run dev

# Client running on port 3000
# Auto-connects to backend API
```

4. Environment Configuration (.env)

Configuration File	Variable Name	Default Value	Description
Frontend (.env)	VITE_API_URL	https://intrakraft-backend.vercel.app/api	Backend endpoint (defaults to live Vercel API; can set localhost:5000)
Frontend (.env)	VITE_FIREBASE_API_KEY	Configured in .env	Firebase Client SDK Web API key for Google Sign-In
Frontend (.env)	VITE_FIREBASE_PROJECT_ID	student-acf58	Firebase project identifier for client auth state
Backend (.env)	PORT	5000	Local HTTP listener port for Express REST API
Backend (.env)	CORS_ORIGIN	*	Allowed origin headers for cross-site requests
Backend (.env)	NODE_ENV	development / production	Environment execution mode

5. Assignment Requirements & Testing Guide

#	Core Requirement	Where & How to Verify in the Application
1	Upload Catalogue Excel (.xlsx)	Navigate to Admin -> Catalogues -> Upload. Drag and drop any .xlsx file. Handled via backend POST /api/catalogues/upload.
2	Display Catalogue Products	View parsed products under Admin -> Products or User View -> Catalogue. Shows exact name, category, and extracted sizes.
3	Dynamic Sizes Extraction	Upload sheets with non-standard sizes (e.g., kids sizes 4-5Y, 5-6Y or adult XS..XXL). System extracts sizes dynamically without hardcoding.
4	Add to Cart & Grade Display	Select products with Grade A, B, or C. In User View -> Cart, items are strictly divided into distinct Grade A, Grade B, Grade C sections.
5	Grade-wise Ratio Matrix	Go to Ratio Studio (/admin/ratios/studio). Set independent ratio distributions for Grade A, Grade B, Grade C, and Grade D.
6	Attribute Combination Levels	Configure ratios at Brick, Category, Brick+Neck, Brick+Sleeve, or custom attribute combinations using dynamic selector buttons.
7	Exact Ratio Calculation	Backend applies the Hamilton Largest Remainder Method, ensuring allocated unit quantities sum exactly to 100 with 0% rounding loss.
8	Cascade Deletion Integrity	Delete a catalogue in Admin -> Catalogues. Its associated products are automatically purged from the product list and dashboard.

6. Build & Lint Verification Commands

```
npm run build          # Validates TypeScript compilation and builds vite bundle (0 errors)
npm run lint           # Validates root TypeScript rules (tsc -b) (0 errors)
cd backend && npm run build # Compiles backend TypeScript to dist/ (0 errors)
```